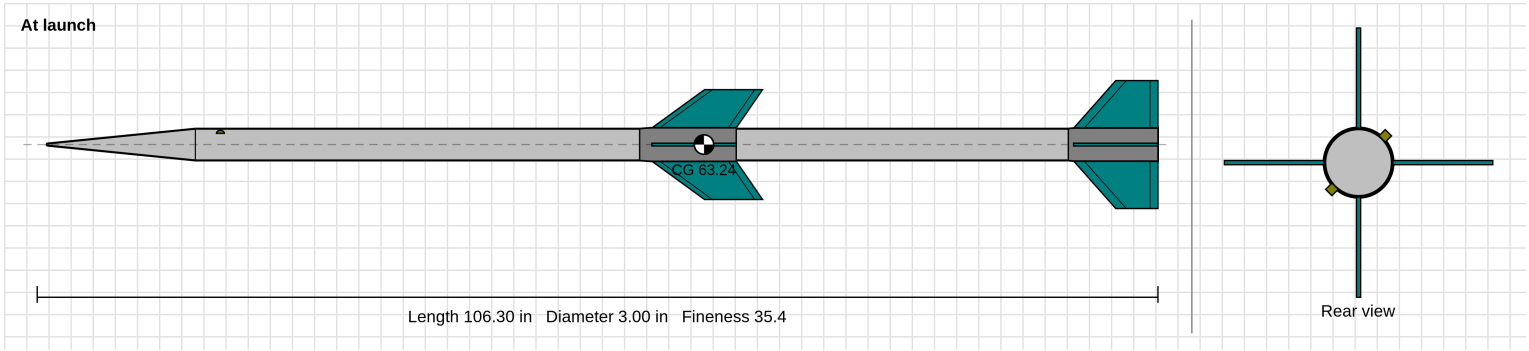
PANDORA - EAYF_CONFIG_1 - Results



PANDORA - EAYF_CONFIG_1
Nose radius for heating: 0.1 in

	nominal	mean	std	-2 sigma	+2 sigma	95% low	95% high	lowest	highest	
Rail exit speed	114.12	114.07	1.2845	111.5	116.64	111.54	116.61	114.66	114.32	ft/s
Rail exit AoA	9.7782	5.6763	3.8925	-2.1086	13.461	-2.0126	13.365	7.039	1.5367	deg
Forecast gust	26.903	14.851	10.246	-5.6401	35.342	-5.3874	35.089	18.045	2.2966	ft/s
Rail exit AoA in gust	13.265	7.3557	5.0302	-2.7046	17.416	-2.5805	17.292	8.9462	1.151	deg
Apogee	1.1731e+05	1.2052e+05	6274.4	1.0797e+05	1.3307e+05	1.0812e+05	1.3291e+05	1.061e+05	1.359e+05	ft
Time to apogee	88.317	89.665	2.7104	84.244	95.086	84.311	95.019	83.369	96.051	s
Max speed	4403.1	4409.7	100.67	4208.4	4611.1	4210.9	4608.6	4171.9	4590.6	ft/s
Max acceleration	74.251	74.347	2.1194	70.108	78.586	70.161	78.533	69.251	77.967	g
Max Mach	4.1642	4.1722	0.098117	3.9759	4.3684	3.9783	4.366	3.9418	4.3523	
Max dynamic pressure	11328	11336	504.54	10327	12346	10340	12333	10228	12105	psf
Peak heat flux	567.61	570.13	38.291	493.55	646.71	494.49	645.77	484.42	637.25	BTU/ft²/s
Heat load	3038.1	3053.4	186.86	2679.7	3427.1	2684.3	3422.5	2593.7	3438.7	BTU/ft²
Tilt at ignition	9.3339	4.8836	3.2475	-1.6114	11.379	-1.5313	11.299	5.8092	2.2742	deg
Altitude at ignition	12333	12350	465.97	11418	13282	11430	13271	12394	12527	ft
Flight time	369.18	372.14	7.2986	357.54	386.74	357.72	386.56	357.26	386.22	s
Landing east	-3.5667	-1.8699	2.1109	-6.0917	2.3519	-6.0396	2.2998	-1.2223	-0.26793	mi
Landing north	-6.5704	-3.3473	2.8754	-9.0981	2.4035	-9.0272	2.3326	-3.3915	-2.0293	mi
Impact speed	44.619	42.359	2.0945	38.17	46.548	38.222	46.497	42.445	40.249	ft/s
Launch mass	38.822	lb, CG 63.24 in from the nose tip								
Min static margin	3.2223	cal (while loaded)								
Max angle of attack	0.72361	deg (while loaded)								

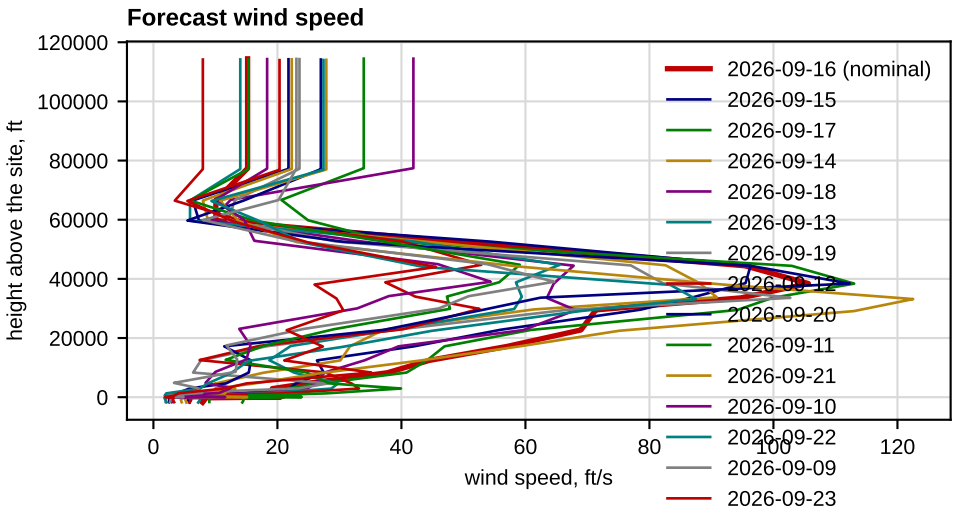
Sustainer ignited: yes
  Monte Carlo: 128 of 128 ignited
Monte Carlo runs: 128
  -2 sigma and +2 sigma are the mean less and plus two standard deviations.
  95% low and high: the prediction interval, where 95% of flights fall: mean -/+ z s sqrt(1 + 1/N),
  z = 1.960, N = 128 runs, assuming a normal spread.
  lowest and highest are the runs with the lowest and highest apogee (runs #1 and #19).
Launch cylinder, 7 nmi: the nominal flight lands inside, 117 of 128 Monte Carlo runs land inside

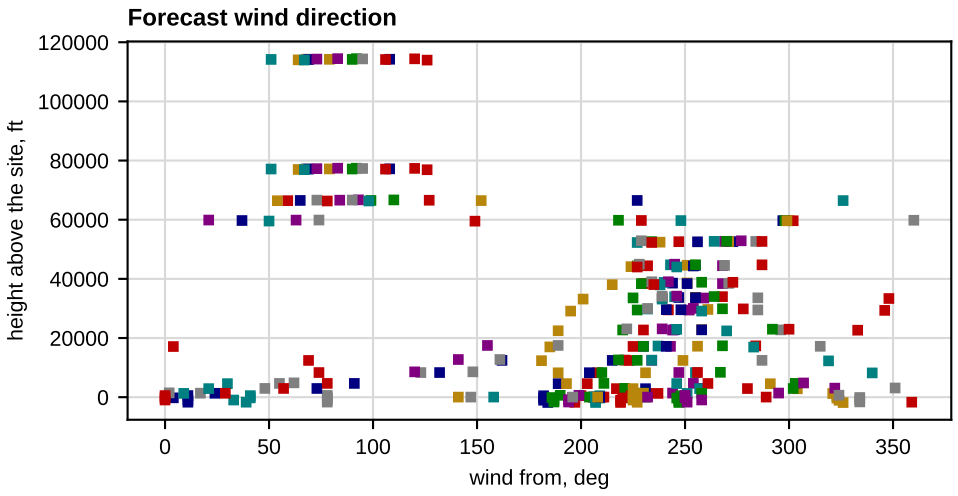
Events (nominal flight)
  2.49 s Booster burnout, separation
 12.49 s Sustainer ignition
 15.85 s Sustainer burnout
 88.32 s Apogee
 88.32 s Apogee chute
321.64 s Main chute

PANDORA - EAYF_CONFIG_1 - Weather and Monte Carlo inputs

```
Weather
  Open-Meteo forecast at 35.2050, -117.4832, 06:00 America/Los_Angeles
  day                   surface wind          gust at the launch hour   runs
  2026-09-16      7.9 ft/s from 197 deg   26.9 ft/s from 219 deg at 06:00      5   nominal flight
  2026-09-15      5.8 ft/s from 184 deg   18.0 ft/s from 210 deg at 06:00      7
  2026-09-17      9.0 ft/s from 187 deg   27.6 ft/s from 211 deg at 06:00     13
  2026-09-14      5.3 ft/s from 326 deg    9.5 ft/s from 327 deg at 06:00     10
  2026-09-18      5.9 ft/s from 194 deg   15.7 ft/s from 221 deg at 06:00      8
  2026-09-13      7.2 ft/s from 207 deg   17.1 ft/s from 219 deg at 06:00      6
  2026-09-19      2.0 ft/s from  78 deg    3.9 ft/s from  62 deg at 06:00      9
  2026-09-12      5.8 ft/s from 219 deg   24.6 ft/s from 217 deg at 06:00     16
  2026-09-20      2.5 ft/s from  11 deg    2.3 ft/s from  39 deg at 06:00     15
  2026-09-11     14.3 ft/s from 247 deg   33.8 ft/s from 241 deg at 06:00      8
  2026-09-21      4.6 ft/s from 227 deg   12.5 ft/s from 219 deg at 06:00      3
  2026-09-10      3.0 ft/s from 251 deg   13.1 ft/s from 277 deg at 06:00      7
  2026-09-22      2.1 ft/s from  39 deg    2.3 ft/s from  47 deg at 06:00      8
  2026-09-09      2.7 ft/s from 334 deg    7.2 ft/s from  20 deg at 06:00      9
  2026-09-23      3.3 ft/s from 359 deg    3.0 ft/s from  14 deg at 06:00      4
  Each day is a whole forecast wind profile; the gust is the 10 m forecast for the launch hour, used only for the rail exit AoA in gust.

Monte Carlo inputs: these 8 change from run to run, everything else is fixed
                           nominal     sigma      lowest      mean   highest   (as drawn)
  Booster impulse              100         2       95.47     99.84     106.1  %
  Sustainer impulse            100         2       93.17     99.94     106.4  %
  Booster dry mass             100         2       94.84     99.68     106.6  %
  Sustainer dry mass           100         2       94.84     99.84     104.5  %
  Coast before ignition         10       0.5       8.691      9.99      11.3  s
  Launch angle                  90       0.5       88.72      89.6        90  deg
  Heading                        0       180      -172.7     5.624     179.5  deg
  Forecast day            2026-09-16   one of the 15 days above, picked at random
  Random seed: different every time
```





PANDORA - EAYF_CONFIG_1 - Inputs

```
[Case]
  Units              IN
  Analyze            STACK

[Nose]
  Shape              CONICAL
  Length             15
  PowerExponent      0.5
  TipRadius          0.1
  FlatTip            false

[Body]
  Diameter           3
  Length             51.3
  BoattailLength     0
  BoattailDiameter   3
  NozzleExitDiameter 2.5

[Fins]
  Count              4
  RootChord          8
  TipChord           5.5
  Span               3.62
  Sweep              5
  AftOffset          0
  Thickness          0.157
  Section            CHAMFER
  LEChamfer          0.75
  TEChamfer          0.75

[FinCan]
  Enabled            true
  Diameter           3.15
  Length             9.16
  AftOffset          0
  FrontBevel         1.09

[Booster]
  Enabled            true
  Diameter           3
  Length             40
  Insertion          0
  TransitionLength   1
  NoseCone           false
  NoseShape          OGIVE
  NoseLength         8
  BoattailLength     0
  BoattailDiameter   0.01
  NozzleExitDiameter 2.75

[BoosterFins]
  Count              4
  RootChord          8
  TipChord           4
  Span               4.5
  Sweep              4
  AftOffset          0
  Thickness          0.118
  Section            CHAMFER
  LEChamfer          1
  TEChamfer          0.75

[BoosterFinCan]
  Enabled            true
  Diameter           3.1
  Length             8.5
  AftOffset          0
  FrontBevel         0.5

[Protuberance1]
  Type               FAIRING
  Stage              SUSTAINER
  X                  17
  Count              2
  Length             0.75
  Width              0.4
  Height             0.4
  Standoff           0

[Flight]
  Altitude           4000
  ClipFinsToShock    false
  TurbulentBoundaryLayerfalse

[Motors]
  Sustainer          EAYF Sustainer - N3051
  Booster            EAYF Booster - M3140.e

[Site]
  Weather            OPENMETEO
  Latitude           35.205
  Longitude          -117.483
  Elevation          2000
  Date               2026-09-16
  Time               06:00
  Timezone           America/Los_Angeles
  WindSpeed          0
  WindDirection      0
  CylinderRadius     7

[Launch]
  RailLength         15
  Angle              90
  Heading            0
  Coast              10
  MaxTilt            15
  MinIgnitionAltitude 0
  MaxTime            1000

[Recovery]
  ApogeeChuteDiameter 8
  ApogeeChuteCd      0.8
  MainChuteDiameter  28
  MainChuteCd        1.8
  MainChuteAltitude  2000

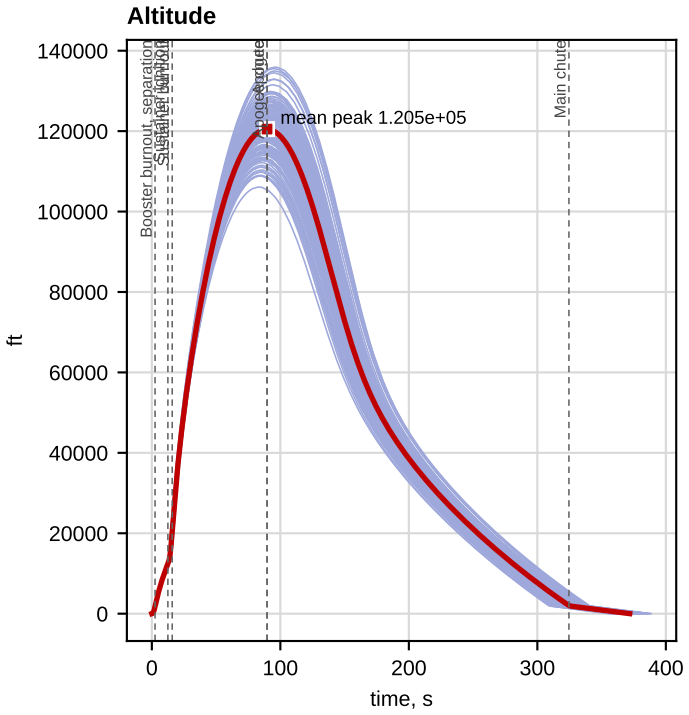
[Thermal]
  IncludeDescent     false

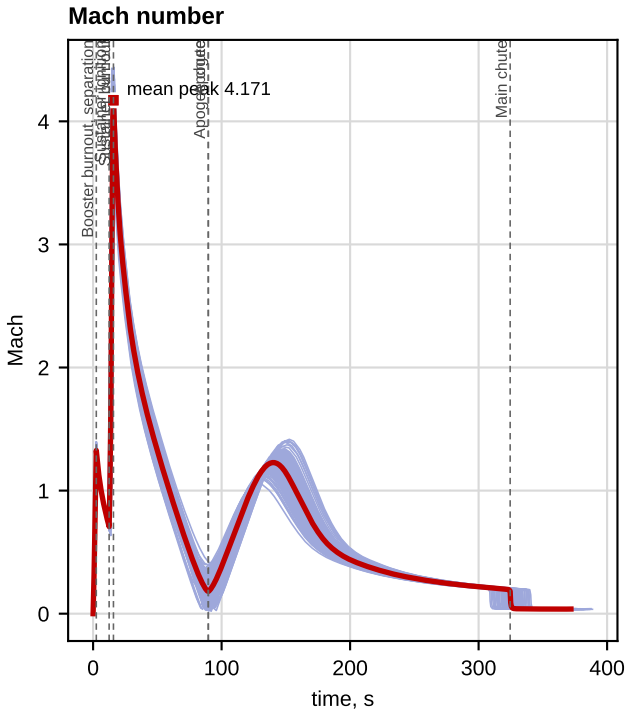
[DrySustainer]
  Mass               13.5
  CG                 42
  Inertia            2321
  MotorOffset        2.6

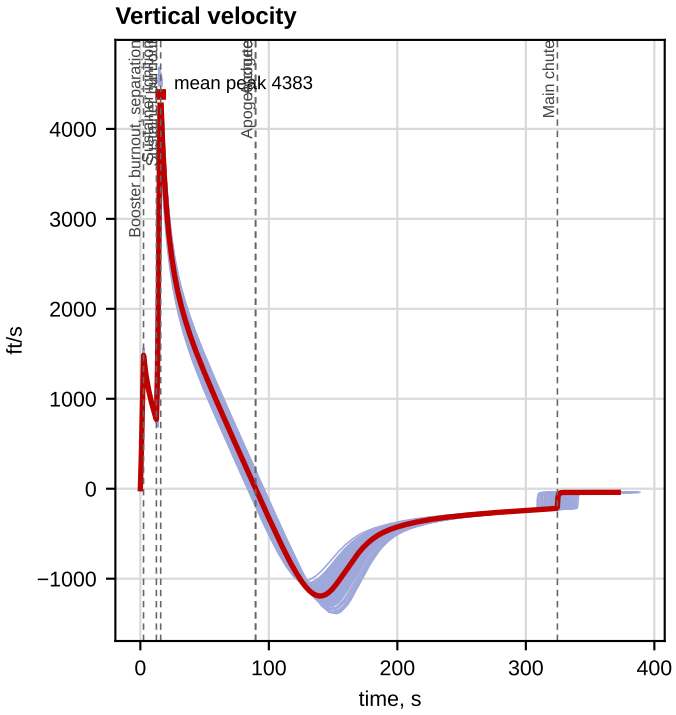
[DryStack]
  Mass               21.62
  CG                 67
  Inertia            10692.5
  MotorOffset        3.44

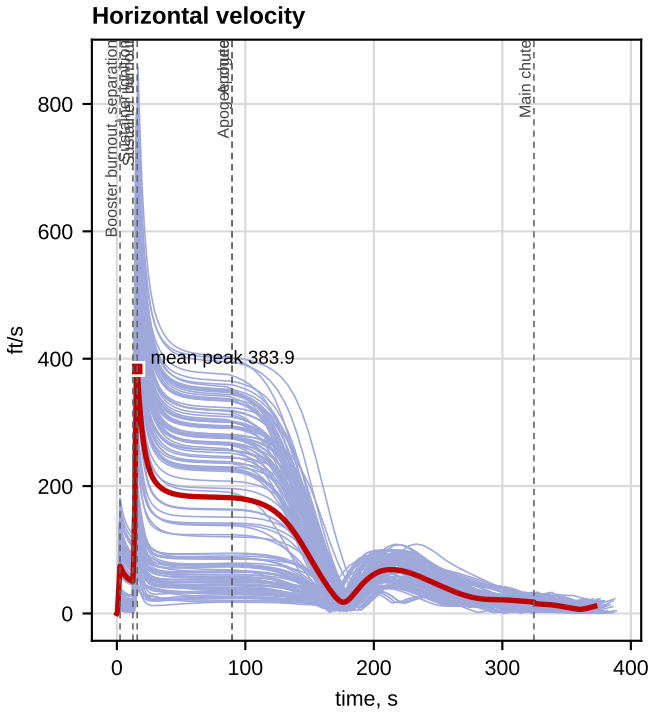
[MonteCarlo]
  Runs               128
  Seed               0
  Prediction         95
  AngleStd           0.5
  HeadingStd         180
  CoastStd           0.5
  BoosterImpulseStd  2
  SustainerImpulseStd 2
  BoosterMassStd     2
  SustainerMassStd   2
  WindSpeedStd       0
  WindDirectionStd   0
  WindDays           7
  MaxDayWind         10
  SkipDays
```

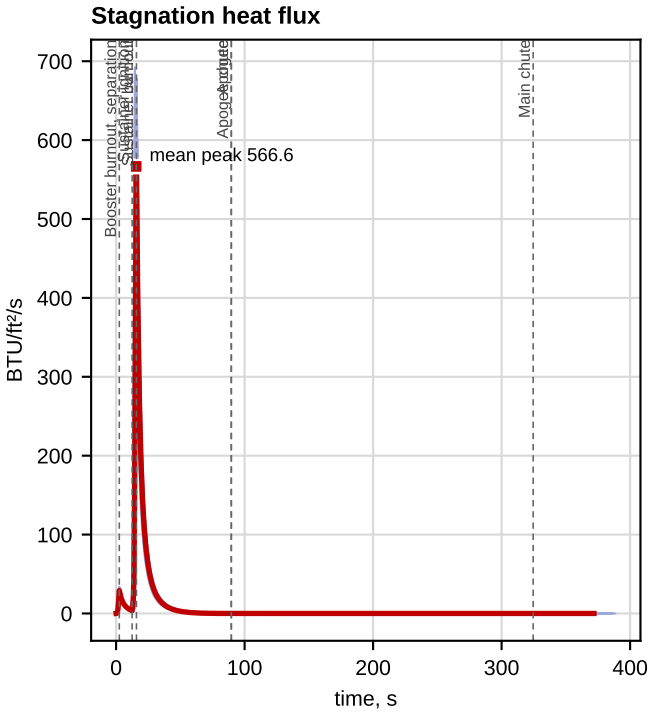
EAYF_CONFIG_1 - Flight, mean of 128 Monte Carlo runs

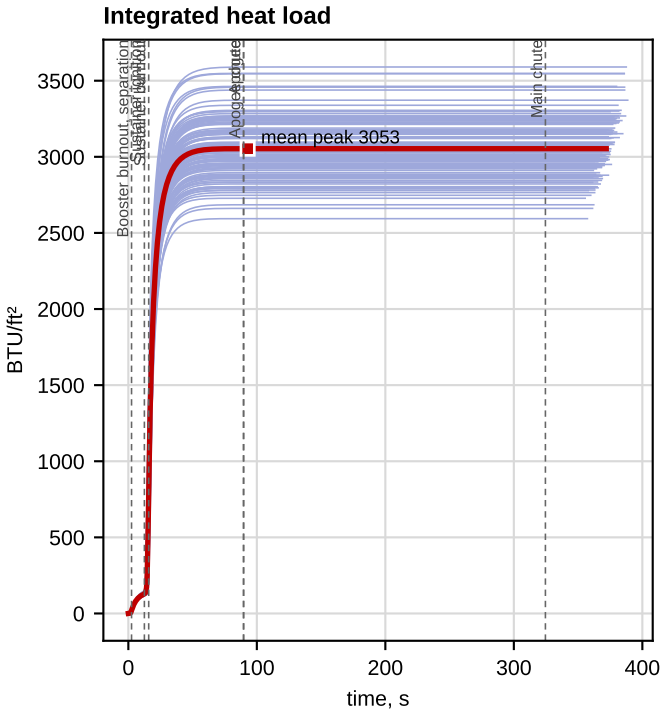






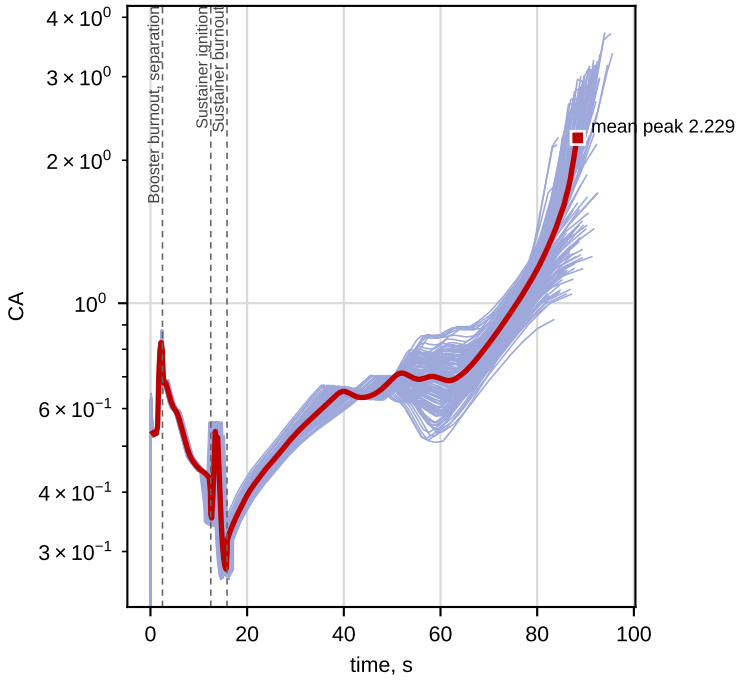




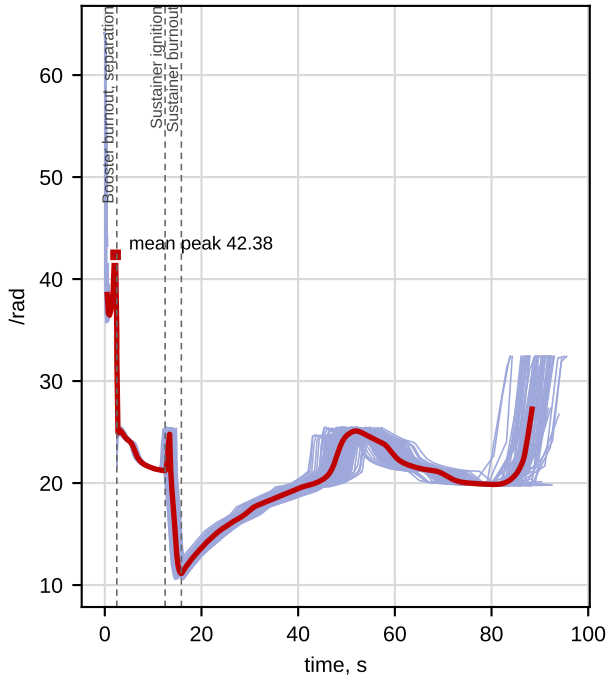


EAYF_CONFIG_1 - DATCOM aerodynamics along the flight (launch to apogee, Mach 0.1 and up)

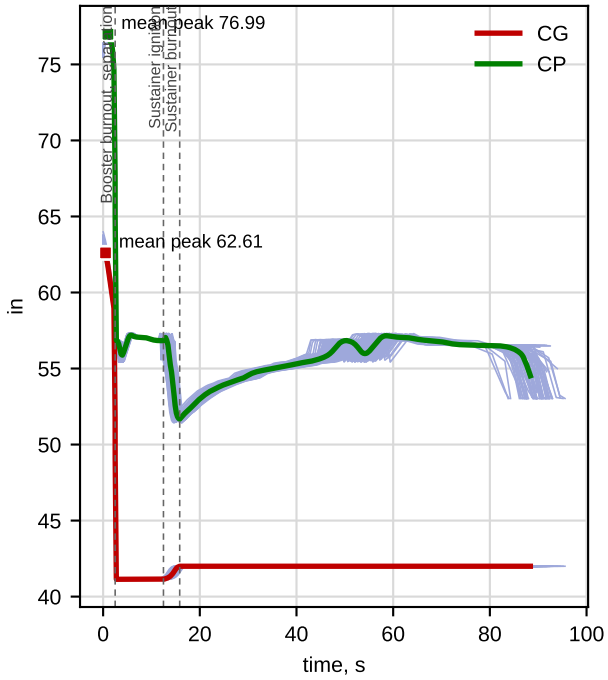
Axial force coefficient CA (log scale)



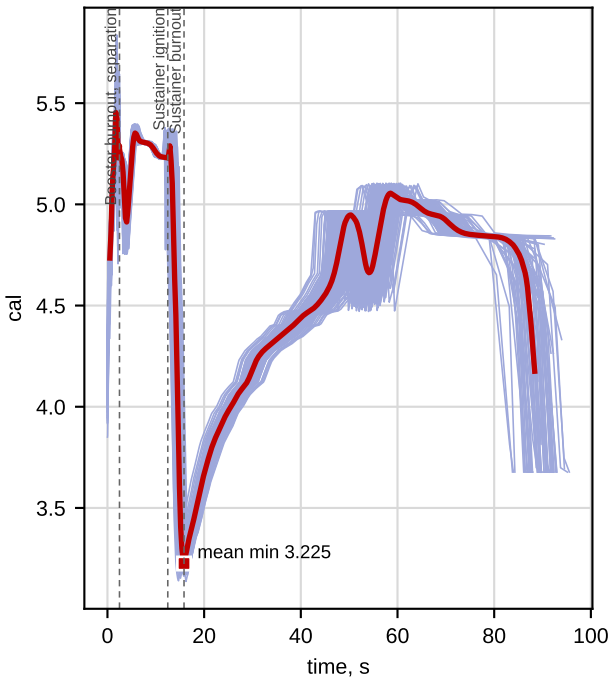
Normal force slope CNα



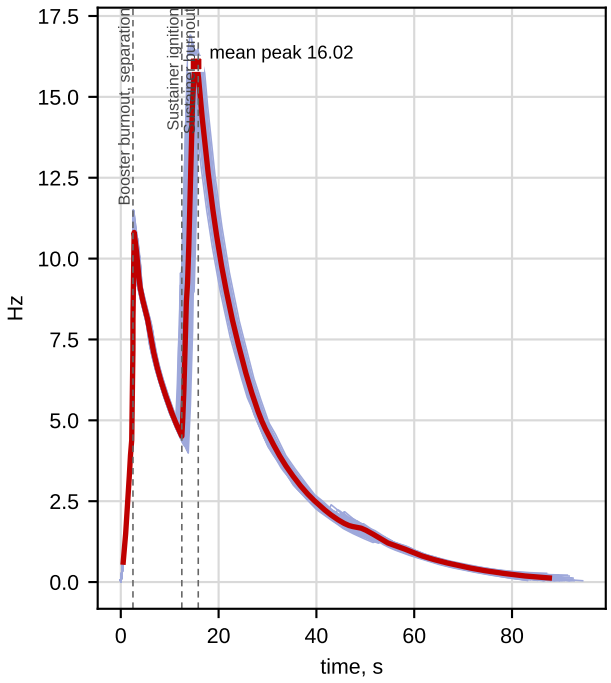
CG and centre of pressure



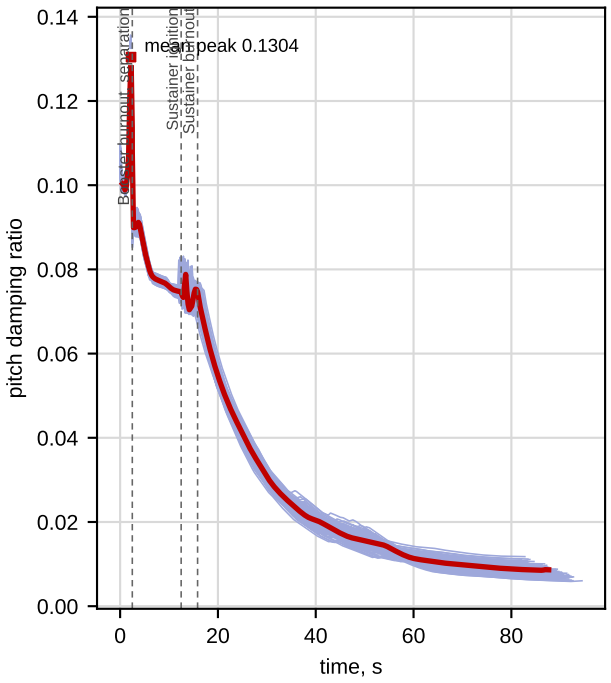
Static margin



Pitch natural frequency

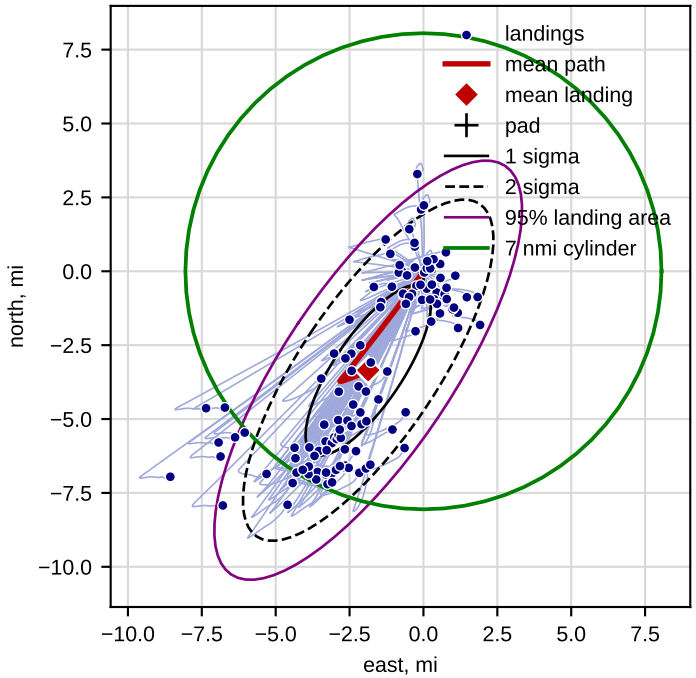


Pitch damping ratio

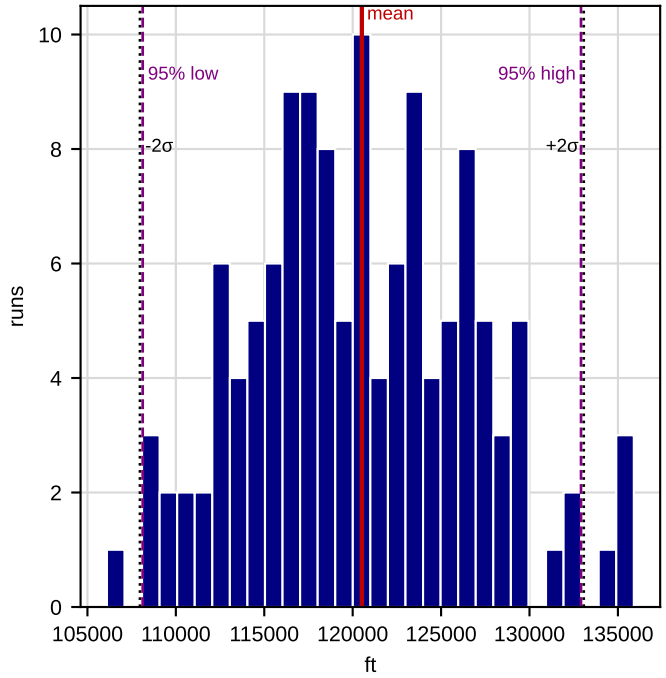


EAYF_CONFIG_1 - Monte Carlo, 128 runs

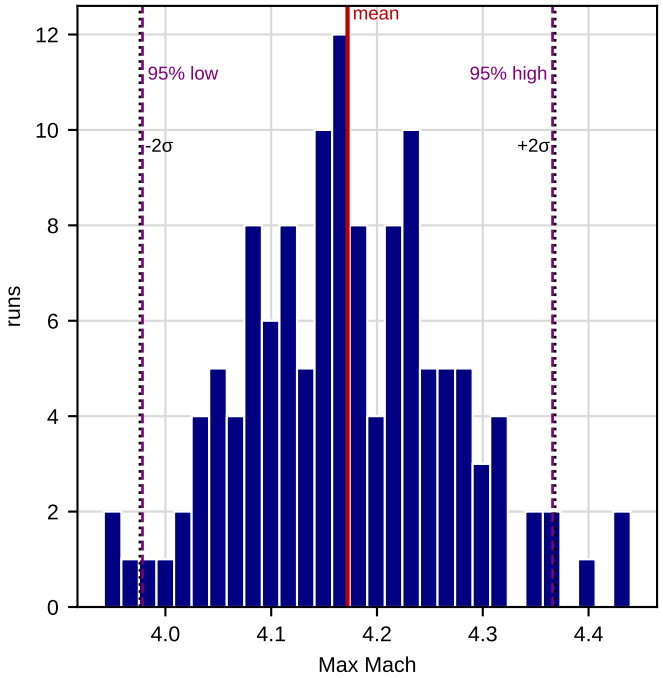
Ground tracks and landing points



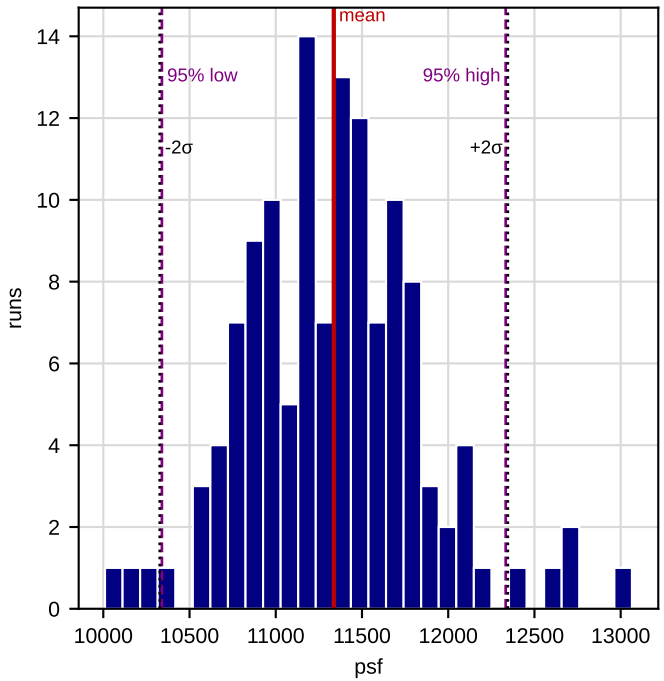
Apogee



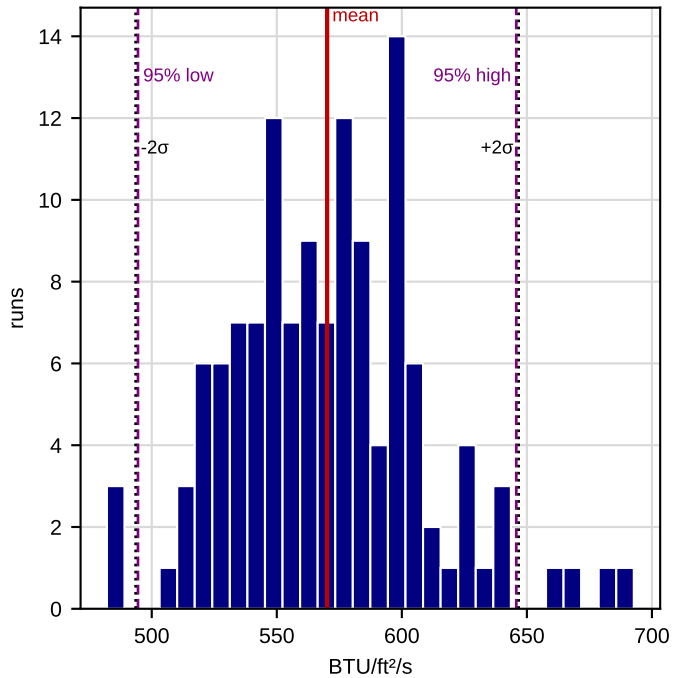
Max Mach



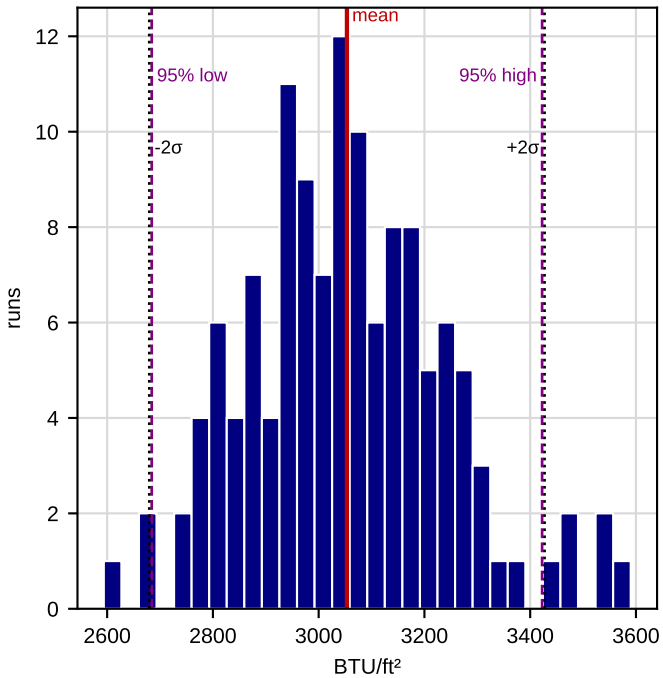
Max dynamic pressure



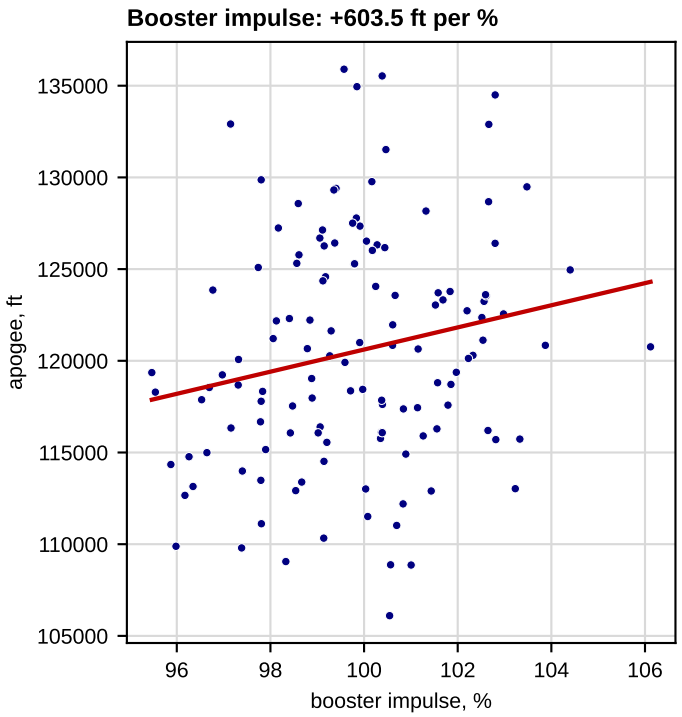
Peak heat flux

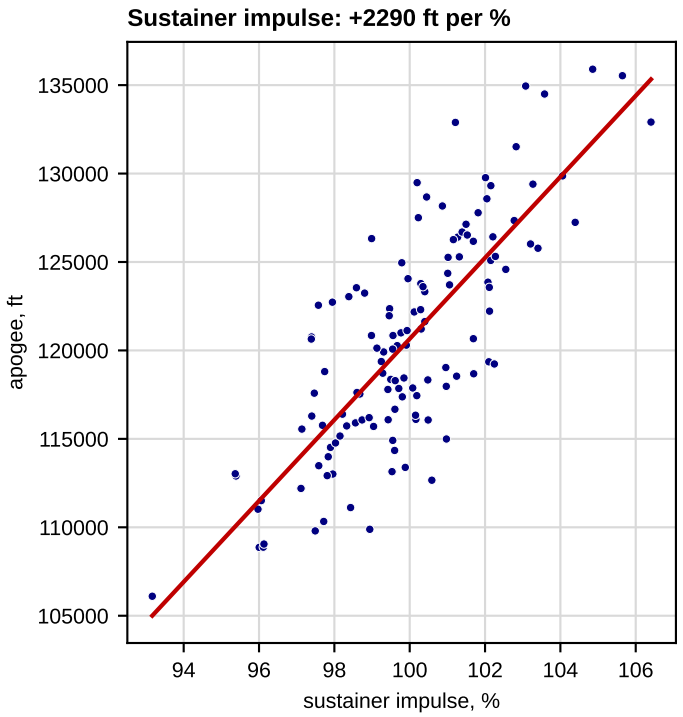


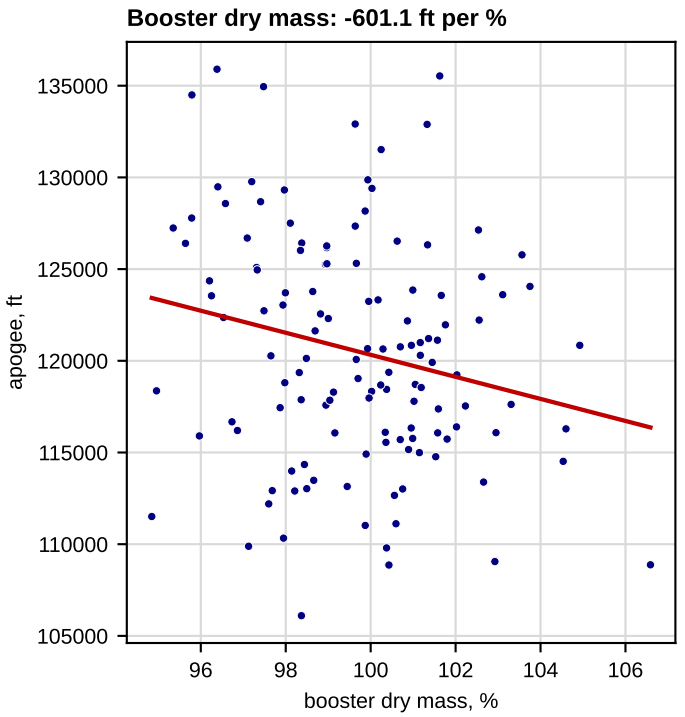
Heat load

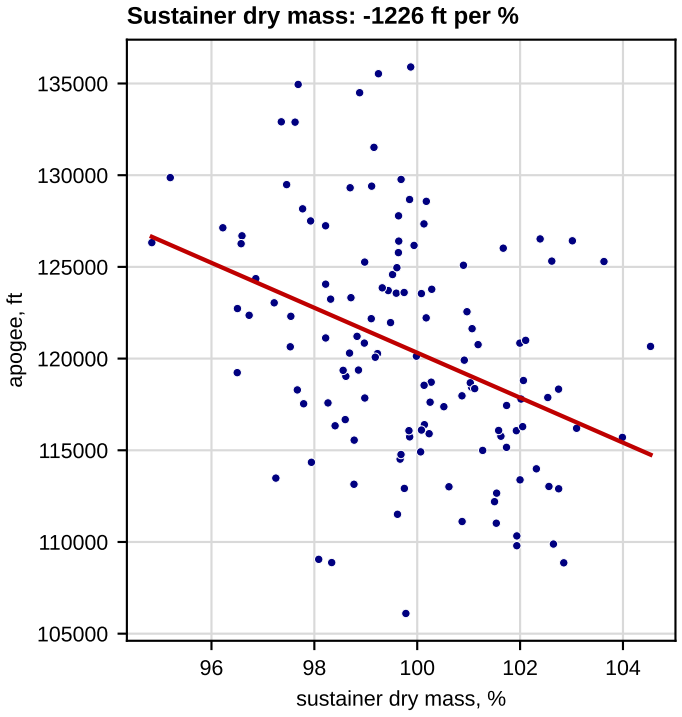


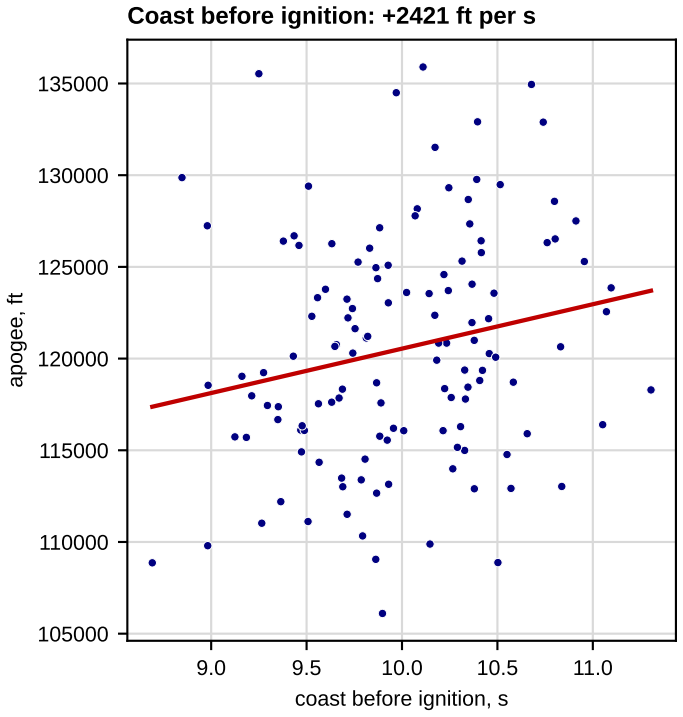
EAYF_CONFIG_1 - Sensitivity: what each varied input did to apogee, 128 runs

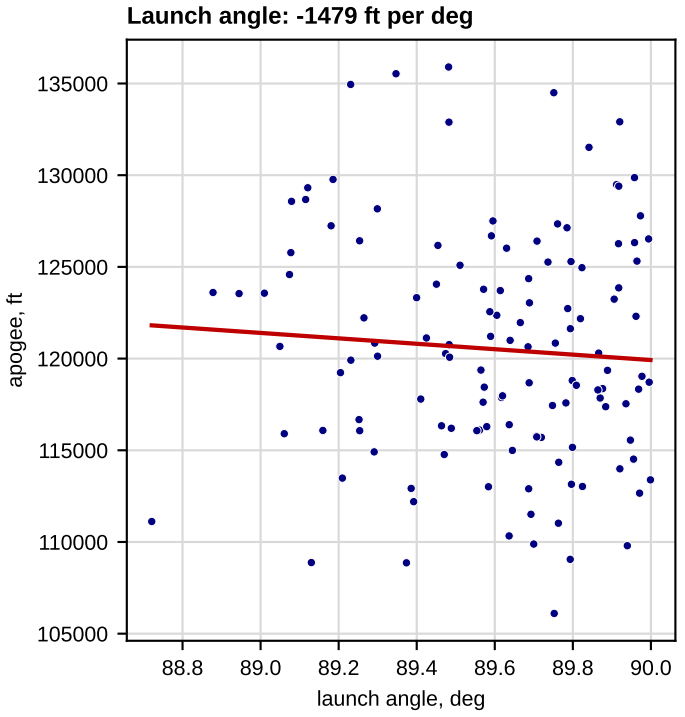












EAYF_CONFIG_1 - Ground track, 128 Monte Carlo runs

